

Decent Coaching Classes

VIII SB

Maths Worksheet

Chapters 4 to 7 | Altitudes & Medians of a Triangle • Expansion Formulae • Factorisation of Algebraic Expressions • Variation

Chapter 4 — Altitudes and Medians of a Triangle

1. Define the median of a triangle.
2. Define the altitude of a triangle.
3. What is the centroid of a triangle? Which symbol is generally used to denote it?
4. State the ratio in which the centroid divides each median, starting from the vertex.
5. In $\triangle ABC$, D is the midpoint of side BC and A, D are joined. Name the segment AD and state which special segment it represents.
6. Draw a rough figure of $\triangle PQR$ and show the altitude drawn from vertex P to side QR.
7. In $\triangle ABC$, the medians AD, BE and CF intersect at the centroid G. If $AG = 8$ cm, find the lengths of GD and AD.
8. In $\triangle XYZ$, XG is a part of median XM, where G is the centroid. If $XM = 15$ cm, find XG and GM.
9. Is the altitude of a triangle always inside the triangle? Explain your answer with reference to an obtuse-angled triangle.
10. Draw $\triangle ABC$ and construct all three medians using a ruler and compass. Label the point where they meet.

Chapter 5 — Expansion Formulae

11. State the formulae for $(a + b)^2$ and $(a - b)^2$.
12. Expand: $(2x + 3y)^2$
13. Expand: $(5a - 4b)^2$
14. Simplify using formulae: $(a + b)^2 - (a - b)^2$
15. Using a suitable formula, find the value of 103^2 .
16. Expand: $(x + 2y + 3z)^2$
17. Using a suitable formula, find the value of 97×103 .
18. Expand: $(2x + 3)^3$
19. Expand: $(a - 2b)^3$
20. If $x + 1/x = 5$, find the value of $x^2 + 1/x^2$.

Chapter 6 — Factorisation of Algebraic Expressions

21. Factorise (take out the common factor): $5x + 15$
22. Factorise: $3x^2 - 6x$
23. Factorise by grouping: $ax + ay + bx + by$
24. Factorise using a standard identity: $x^2 - 25$
25. Factorise: $9a^2 - 16b^2$
26. Factorise: $x^2 + 8x + 16$
27. Factorise: $4x^2 - 12x + 9$

28. Factorise the quadratic trinomial: $x^2 + 7x + 12$

29. Factorise: $x^2 - 5x - 24$

30. Factorise: $2x^2 + 11x + 5$

Chapter 7 — Variation

31. Define direct variation and give one real-life example.

32. Define inverse variation and give one real-life example.

33. If y varies directly as x , and $y = 15$ when $x = 5$, find y when $x = 8$.

34. If y varies inversely as x , and $y = 4$ when $x = 9$, find y when $x = 6$.

35. The cost of apples varies directly as their number. If 6 apples cost ₹90, find the cost of 10 apples.

36. The number of days needed to finish a piece of work varies inversely as the number of workers. If 8 workers complete the work in 15 days, in how many days will 12 workers complete it?

37. If a varies directly as the square of b , and $a = 20$ when $b = 2$, find a when $b = 5$.

38. For a fixed distance, speed varies inversely as time taken. A car covers the distance at 60 km/hr in 3 hours. Find its speed if it takes 2 hours to cover the same distance.

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Answer Key & Solutions

Chapter 4 — Altitudes and Medians of a Triangle

1. A median of a triangle is the line segment joining a vertex to the midpoint of the opposite side.
2. An altitude of a triangle is the perpendicular line segment drawn from a vertex to the line containing the opposite side.
3. The centroid is the point where the three medians of a triangle intersect. It is denoted by G.
4. The centroid divides each median in the ratio 2 : 1, measured from the vertex to the midpoint of the opposite side.
5. AD is called a median of $\triangle ABC$, since D is the midpoint of BC.
6. In an acute-angled $\triangle PQR$, the altitude from P meets QR at a point between Q and R, forming a right angle with QR (figure to be drawn by student).
7. $AG : GD = 2 : 1$. Since $AG = 8$ cm, $GD = 4$ cm. Therefore $AD = AG + GD = 8 + 4 = 12$ cm.
8. $XG : GM = 2 : 1$, so $XG = (2/3) \times XM = (2/3) \times 15 = 10$ cm, and $GM = (1/3) \times 15 = 5$ cm.
9. No. In an obtuse-angled triangle, the altitudes from the two acute-angled vertices fall outside the triangle, on the extension of the opposite side.
10. All three medians of any triangle always meet at a single point — the centroid, G.

Chapter 5 — Expansion Formulae

11. $(a + b)^2 = a^2 + 2ab + b^2$ and $(a - b)^2 = a^2 - 2ab + b^2$
12. $(2x + 3y)^2 = (2x)^2 + 2(2x)(3y) + (3y)^2 = 4x^2 + 12xy + 9y^2$
13. $(5a - 4b)^2 = (5a)^2 - 2(5a)(4b) + (4b)^2 = 25a^2 - 40ab + 16b^2$
14. $(a + b)^2 - (a - b)^2 = (a^2 + 2ab + b^2) - (a^2 - 2ab + b^2) = 4ab$
15. $103^2 = (100 + 3)^2 = 100^2 + 2(100)(3) + 3^2 = 10000 + 600 + 9 = 10609$
16. $(x + 2y + 3z)^2 = x^2 + 4y^2 + 9z^2 + 4xy + 12yz + 6xz$ (using $(a+b+c)^2 = a^2+b^2+c^2+2ab+2bc+2ac$ with $a=x, b=2y, c=3z$)
17. $97 \times 103 = (100 - 3)(100 + 3) = 100^2 - 3^2 = 10000 - 9 = 9991$
18. $(2x + 3)^3 = (2x)^3 + 3(2x)^2(3) + 3(2x)(3)^2 + 3^3 = 8x^3 + 36x^2 + 54x + 27$
19. $(a - 2b)^3 = a^3 - 3(a)^2(2b) + 3(a)(2b)^2 - (2b)^3 = a^3 - 6a^2b + 12ab^2 - 8b^3$
20. $x^2 + 1/x^2 = (x + 1/x)^2 - 2 = 5^2 - 2 = 25 - 2 = 23$

Chapter 6 — Factorisation of Algebraic Expressions

21. $5x + 15 = 5(x + 3)$
22. $3x^2 - 6x = 3x(x - 2)$
23. $ax + ay + bx + by = a(x + y) + b(x + y) = (x + y)(a + b)$
24. $x^2 - 25 = x^2 - 5^2 = (x + 5)(x - 5)$
25. $9a^2 - 16b^2 = (3a)^2 - (4b)^2 = (3a + 4b)(3a - 4b)$
26. $x^2 + 8x + 16 = x^2 + 2(x)(4) + 4^2 = (x + 4)^2$
27. $4x^2 - 12x + 9 = (2x)^2 - 2(2x)(3) + 3^2 = (2x - 3)^2$
28. $x^2 + 7x + 12$: find two numbers with sum 7, product 12 $\rightarrow 3$ and 4. $x^2 + 3x + 4x + 12 = x(x+3) + 4(x+3) = (x + 3)(x + 4)$
29. $x^2 - 5x - 24$: find two numbers with sum -5, product -24 $\rightarrow -8$ and 3. $x^2 - 8x + 3x - 24 = x(x-8) + 3(x-8) = (x - 8)(x + 3)$
30. $2x^2 + 11x + 5$: find two numbers with sum 11, product 10 ($=2 \times 5$) $\rightarrow 10$ and 1. $2x^2 + 10x + x + 5 = 2x(x+5) + 1(x+5) = (x + 5)(2x + 1)$

Chapter 7 — Variation

- 31.** In direct variation, as one quantity increases the other increases in the same proportion ($y = kx$). Example: the cost of sugar varies directly as its weight.
- 32.** In inverse variation, as one quantity increases the other decreases proportionally, so their product stays constant ($xy = k$). Example: the number of workers and the number of days needed to finish a job.
- 33.** $y = kx$. Using $y=15$, $x=5$: $k = 3$. When $x = 8$, $y = 3 \times 8 = 24$.
- 34.** $xy = k$. Using $x=9$, $y=4$: $k = 36$. When $x = 6$, $y = 36 \div 6 = 6$.
- 35.** Cost varies directly as number, so cost/number = k . $k = 90/6 = 15$ per apple. Cost of 10 apples = $15 \times 10 = ₹150$.
- 36.** Workers $\times$ days = k (inverse variation). $k = 8 \times 15 = 120$. For 12 workers, days = $120 \div 12 = 10$ days.
- 37.** $a = kb^2$. Using $a=20$, $b=2$: $20 = k(4)$, so $k = 5$. When $b = 5$, $a = 5 \times 25 = 125$.
- 38.** Speed $\times$ time = k (constant distance). $k = 60 \times 3 = 180$. When time = 2 hrs, speed = $180 \div 2 = 90$ km/hr.